

INTRODUCTION TO THE INTEGRATED LAPLACE APPROXIMATION



1 INLA and latent Gaussian models

The Integrated Nested Laplace Approximation (INLA) is a fast and accurate Bayesian approximation method for the albeit wide class of Latent Gaussian Models (LGMs). This includes GLMs, GLMMs, GAM-like models and a wide range of spatial and spatiotemporal models.

Models of this type can be written as

$$\begin{aligned}\mathbf{y} \mid \mathbf{u}, \theta &\sim \prod_i \pi(y_i \mid \eta_i, \theta), \\ \eta &= A_1 \mathbf{u}_1 + A_2 \mathbf{u}_2 + \dots + A_k \mathbf{u}_k, \\ \mathbf{u} \mid \theta &\sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1}(\theta)), \\ \theta &\sim \pi(\theta).\end{aligned}$$

The first step in defining a LGM within the Bayesian framework is to identify a distribution for the observed data $\mathbf{y} = (y_1, \dots, y_n)$ where typically the mean $\mathbb{E}(y_i) = \mu_i$ of each observation y_i is linked to a linear predictor η_i through an appropriate link function $g(\cdot)$. The additive linear predictor η_i can be defined by covariates (i.e., fixed effects) and different types of random effects:

$$\eta_i = \alpha + \sum_{j=1}^P \beta_j x_{ij} + \sum_{k=1}^L f_k(z_{ik}) \quad i = 1, \dots, n.$$

Here, α is the intercept, $\beta_j, j = 1, \dots, P$, are coefficients of P covariates, and the functions $f_k(\cdot)$ can take different forms such as smooth and nonlinear effects of covariates, time trends and seasonal effects, random intercept and slopes as well as temporal or spatial random effects. We denote the vector of all latent effects as $\mathbf{u}$ and it is assumed to be Gaussian Markov random field (GMRF). This GMRF will have a zero mean and precision matrix $\mathbf{Q}(\theta)$ which consists of sums of the precision matrices of the fixed effects and the other model components. Furthermore, θ represents a vector of hyperparameters that control the behavior of the latent components and the likelihood of observations ($p(\mathbf{y} \mid \mathbf{u}, \theta)$), which we assume are conditionally independent given the latent field $\mathbf{u}$.

The core idea behind the INLA approach is that, rather than estimating the joint posterior distribution of the model parameters $p(\mathbf{u}, \theta \mid \mathbf{y})$, we focus on individual posterior marginals of the model parameters ($p(u_i \mid \mathbf{y})$ and $p(\theta_j \mid \mathbf{y})$). This is achieved thanks to the computational properties of GMRF and the Laplace approximation for multidimensional integration.